Chapter 7 Elements

EXTRA WORKSHEET 2: Mixtures

SCIENCE UNDERSTANDING **VERBAL/LINGUISTIC, VISUAL/SPATIAL**

Name:_____Class:_____Date: _____

A mixture of two substances can be divided into two components:

- the dispersion medium, which makes up most of the mixture
- the dispersed phase, which is the substance that is mixed into the dispersion medium.

For example, when a teaspoon of sugar is dissolved into a cup of water, the water is the dispersion medium and the sugar is the dispersed phase.

Both the dispersion medium and the dispersed phase can be a solid, liquid or gas. For example, when there is a dust storm, the dust is a solid dispersed phase and the air is a gaseous dispersion medium.

Questions

1 Complete the following to **identify** the dispersion medium and dispersed phase for each mixture.

Mixture		Dispersion medium	Dispersed phase
a	salt water		
b	smoke from a fire		
c	muddy water		
d	fog		
e	whipped cream		

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The type of mixture depends on whether the dispersion medium and dispersed phase are solids, liquids or gases. The table below outlines the names given to various types of mixtures.

		Dispersion medium		
		solid	**liquid**	**gas**
Dispersed phase	**solid**	solid solution alloy (for metals) solid sol	solution sol suspension	solid areosol
	liquid	solution gel	liquid solution emulsion	aerosol
	gas	solid foam	solution foam	gas solution

2 Complete the table below to **identify** one example of each type of mixture. The first one has been done for you.

		Dispersion medium		
		solid	**liquid**	**gas**
Dispersed phase	**solid**	e.g. brass alloy		
	liquid			
	gas			

3 **Draw** pictures on the particle level to represent each of the following:

a alloy	**b** aerosol	**c** solid foam

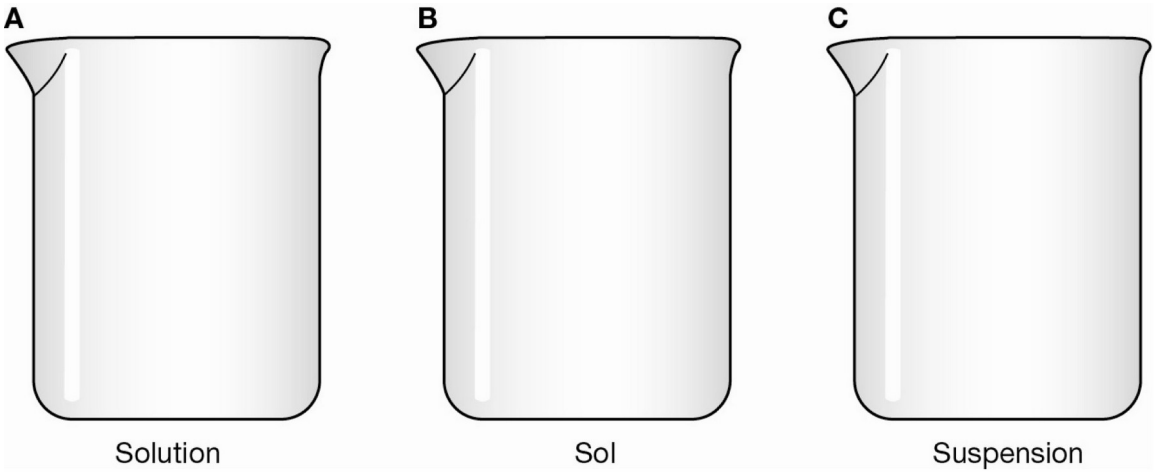
There are three types of mixtures where a solid dispersed phase is mixed into a liquid dispersion medium.

- Solution: solid dispersed phase dissolves into the smallest possible particles, which are invisible to the naked eye

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- Sol: solid dispersed phase is made up of slightly larger particles that are still invisible to the naked eye but can scatter light
- Suspension: solid dispersed phase is made up of even larger particles that may be visible to the naked eye and will settle to the bottom of the dispersion medium if left for long enough

4 **Draw** diagrams representing each of these types of mixtures inside the beakers:



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